## Investigation: HERCULES S.A. FABRICA DE TALHERES (CVM code 6629), fiscal year 2013, as it was on 2024-12-31

Statistical signal based on public financial statements. Not an accusation, not investment advice.

## Company

Sector: Metalurgia e Siderurgia (origin: FCA); group: non-financial.
Before 2018, the company was called HERCULES S/A - FABRICA DE TALHERES.
Before 2011, the company was called HÉRCULES S/A - FÁBRICA DE TALHERES.
Trading codes (2023 FCA, version 1, filed on 2023-02-09): HETA3 (Ações Ordinárias, Bolsa, since 1968-11-01); HETA4 (Ações Preferenciais, Bolsa, since 1968-11-01).

## Document used

2013 DFP, version 1, filed on 2014-04-17; origin: open data; type: individual; original scale: MIL.
Versions known on the date: 1 (2014-04-17).
Document at the CVM: http://www.rad.cvm.gov.br/ENETCONSULTA/frmDownloadDocumento.aspx?CodigoInstituicao=1&NumeroSequencialDocumento=36323

## Possible manipulation (engine v1, study 1)

Beneish M-score
- Value: component not computable: DEPI
- Alert by the study 1 rule: does not trigger alerts
- Rule: M > −1.78 (pre-registration, section 4)
- Percentile in sector and year: —
- Percentile in group and year: —
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: low

DSRI, days' sales in receivables index
- Value: 0.920 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 36 (n = 21)
- Percentile in group and year: 38 (n = 460)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

GMI, gross margin index
- Value: 1.000 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 69 (n = 21)
- Percentile in group and year: 50 (n = 478)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

AQI, asset quality index
- Value: 1.032 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 55 (n = 21)
- Percentile in group and year: 60 (n = 474)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

SGI, sales growth index
- Value: 0.900 (index)

M-score component; it has no rule of its own
Percentile in sector and year: 17 (n = 21)
Percentile in group and year: 14 (n = 480)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

DEPI, depreciation index
Value: zero denominator: DEPI, t−1 depreciation + property, plant and equipment
M-score component; it has no rule of its own
Percentile in sector and year: —
Percentile in group and year: —
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: low

SGAI, SG&A expenses index
Value: 1.465 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 93 (n = 21)
Percentile in group and year: 92 (n = 471)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

TATA, total accruals to total assets
Value: −5.767 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 2 (n = 21)
Percentile in group and year: 1 (n = 501)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

LVGI, leverage index
Value: 101.006 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 98 (n = 21)
Percentile in group and year: 100 (n = 495)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

DPO jump
Value: zero denominator: DPO, t cost of sales
Alert by the study 1 rule: does not trigger alerts
Rule: DPO change in the year's top decile among non-financial companies (≥ 90th percentile of computable values on the date: 26.5 days)
Percentile in sector and year: —
Percentile in group and year: —
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: low
Alternative explanations: Bargaining power with suppliers and a change in purchase mix (decision of 2026-11-07)

Implied interest rate minus average CDI
Value: 13.71 p.p.
Alert by the study 1 rule: yes
Rule: implied interest rate above the year's average CDI rate + 10 p.p. (pre-registration, section 4)
Percentile in sector and year: 69 (n = 21)

Percentile in group and year: 72 (n = 444)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium
Alternative explanations: Expensive debt, exchange differences and lease interest inside finance costs (decision of 2026-11-07)

Accruals to average total assets
Value: −5.530 (index)
Alert by the study 1 rule: does not trigger alerts
Percentile in sector and year: 2 (n = 21)
Percentile in group and year: 1 (n = 500)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

## Financial risk (context only)

Altman Z'' (financial risk, context only)
Value: −243.122 (index)
Components: X1 = −0.777; X2 = −73.922; X3 = 0.594; X4 = −0.986 (indices)
Percentile in sector and year: 2 (n = 22)
The company's own history: fewer than 5 previous fiscal years with a value
- No cut-off zone: the available source does not define zones for Z''
- Coefficients estimated outside Brazil; bankruptcy models lose validity over time (Pereira and Martins 2015, p. 25)

## Financial quality (context only)

Piotroski F (financial quality, context only)
Value: F = 2/9 (8 computable)
Signals:
- Return on assets (ROA): 0
- Operating cash flow on assets (CFO): 1
- Change in ROA: 0
- Cash above earnings (ACCRUAL): 1
- Change in leverage: 0
- Change in current ratio: 0
- No equity issuance (EQ_OFFER): —
- Change in gross margin: 0
- Change in asset turnover: 0
- No equity issuance (EQ_OFFER), from the cash flow statement: 6.03 lines of the frozen "emissões" pattern (reconnaissance, part 4.2), only those about shares or capital, as the article defines it ("did not issue common equity", p. 9); adaptation: preferred shares also count, because in Brazil they are equity, and the line is identified by text in the cash flow statement

## Changes

Restatement through the following year's comparative: no difference above the tolerance, or the next year's DFP did not exist yet on the date.

## What was known on the date

- DFP 2010, v1: 2011-04-28
- DFP 2011, v1: 2012-04-10
- DFP 2011, v2: 2012-04-11
- DFP 2011, v3: 2012-04-21
- FCA 2013, v1: 2013-04-15
- DFP 2012, v1: 2013-04-23

- FCA 2013, v2: 2013-05-30
- FCA 2013, v3: 2013-06-17
- DFP 2013, v1: 2014-04-17
- DFP 2014, v1: 2015-04-30

DFP and FCA filed with the CVM by the reference date; ITR comes in a later phase.
Material facts and market notices filed by the date (IPE open data, since 2010): 4 material fact documents and 39 market notice documents.
Material facts filed in the 24 months before the date (from 2023-01-01 to 2024-12-31):
- 2023-02-25: Acordo de transação/Parcelamento

Older material facts, the market notices and the link to each document are in the JSON and on screen.
Material facts and market notices filed with the CVM by the date, from the IPE open data (since 2010): filing date, category, subject (as filed, in Portuguese) and link. The content of the documents is not downloaded, read or summarized, and the list generates no alert, score or ranking. The counts are of documents: versions, resubmissions and English versions count separately; in the list, versions of the same document and documents with the same date, category and subject appear on a single line, with a link to each one.

## Data quality

53 applicable checks; 0 failed.

## Auditor's report and auditor (factual section; no alert)

Auditor's report: Com Ressalva (qualified).
Excerpt checked literally: "Em nossa opinião, exceto pelos possíveis efeitos, se houver, do assunto mencionado no parágrafo Base para opinião com ressalva, as demonstrações financeiras acima referidas apresentam adequadamente, em todos os aspectos relevantes, a posição patrimonial e financeira da Hercules S.A. - Fábrica de Talheres em 31 de dezembro de 2013, o desempenho de suas operações e os seus fluxos de caixa para o exercício findo naquela data, de acordo com as práticas contábeis adotadas no Brasil."
Auditor in the 2021 FCA: Baker Tilly Brasil RS Auditores Independentes Sociedade Simples.
Auditor in the 2022 FCA: Baker Tilly Brasil RS Auditores Independentes Sociedade Simples, Taticca Auditores Independentes S.S.
Auditor in the 2023 FCA: Baker Tilly Brasil RS Auditores Independentes Sociedade Simples, Taticca Auditores Independentes S.S.
Auditor changes (one line per change, canonical name by the firm's CVM code):
- 2012: Grant Thornton Auditores Independentes -> KPMG Auditores Independentes. The outgoing firm audited 3 consecutive fiscal years (2009 to 2011), by the start and end dates in the FCA.
- 2017: KPMG Auditores Independentes -> Baker Tilly Brasil RS Auditores Independentes Sociedade Simples (transition: two firms in the 2017 FCA). The outgoing firm audited 5 consecutive fiscal years (2012 to 2016), by the start and end dates in the FCA. The change coincides with the end of the 5-fiscal-year term and is compatible with the rotation rule (art. 31 of CVM Instruction No. 308/1999, in force when the new firm started).
- 2022: entry of Taticca Auditores Independentes S.S, with no other firm leaving.

Auditor rotation: under CVM Resolution No. 23/2021, art. 31, the same independent auditor may not serve the same client for more than five consecutive fiscal years, with a minimum interval of three fiscal years before being hired again; under art. 31-A, the term is up to ten consecutive fiscal years when the company has a statutory audit committee in permanent operation and the auditor is a firm. The Resolution has been in force since 2021-04-01 (art. 41); before that, the equivalent rule was in arts. 31 and 31-A of CVM Instruction No. 308/1999, revoked by the Resolution (art. 40, I). A change of auditor is a fact, not an alert.

## Document excerpts (copied; no alert)

Excerpts copied from the documents, with the page, and the Reference Form table. It is not a signal family: it generates no alert, score or ranking. Absence of mention does not prove absence.
How the reader was tested: 20 held-out cases, a single measurement with the frozen reader

(c99c953): precision 0.962, coverage 0.962, literal check on 132 of 132 excerpts, no natural-person name (docs/leitor_medida.json).

Explanatory notes · Related parties
Note: 11 Partes relacionadas (pages 11 to 11)
Fields of item 18 of CPC 05 (R1)
- [balances] “Saldo ativo por mútuo e conta corrente - - - 261 261” (p. 11, original excerpt in Portuguese)
- [balances] “Saldo passivo por mútuo e conta corrente - 2.966 - - 2.966” (p. 11, original excerpt in Portuguese)
- [amount of the transactions] “Despesas financeiras 41.154 - - - 41.154” (p. 11, original excerpt in Portuguese)
- [amount of the transactions] “Em 31 de dezembro de 2012” (p. 11, original excerpt in Portuguese)
- [balances; amount of the transactions] “Saldo ativo por mútuo e conta corrente - - 623 261 884” (p. 11, original excerpt in Portuguese)
- [balances; amount of the transactions] “Saldo passivo por mútuo e conta corrente 349.040 2.966 - - 352.006” (p. 11, original excerpt in Portuguese)
- [amount of the transactions] “Despesas financeiras 37.214 - - - 37.214” (p. 11, original excerpt in Portuguese)
- [terms and conditions] “Empresa Remuneração Vencimento Bellini S.A. Não remunerada Indeterminado Laboratorio Avamiller de Cosméticos Ltda. Não remunerada Indeterminado” (p. 11, original excerpt in Portuguese)
- [amount of the transactions] “Em 13 de dezembro de 2013 a Companhia emitiu debêntures no montante de R$ 389.007, que foram” (p. 11, original excerpt in Portuguese)
- [balances] “subscritas no seu total avista pelo saldo decorrente conta de mútuo por conta corrente, conforme descrito na nota explicativa 12.” (p. 11, original excerpt in Portuguese)

Explanatory notes · Reverse factoring and forfaiting
The reader found no mention of reverse factoring (risco sacado), forfaiting or supplier agreements in the notes of this DFP. Absence of mention does not prove absence of the operation.

Auditor's report
Type of opinion: qualified
- [basis for the opinion] “Base para opinião com ressalva Conforme mencionado na nota explicativa 9, a Companhia mantém saldos contábeis passivos e de resultado relativos às suas obrigações fiscais, incluindo obrigações sociais. Os valores apresentados pelos órgãos fiscais, relativos aos impostos a recolher apresentam divergências em relação aos saldos contábeis da Companhia. A Companhia está em processo de revisão dos saldos tributários divergentes com as posições dos órgãos fiscais e outras apurações dos valores devidos, que envolvem um volume significativo de processos, informações de terceiros e estudos da legislação para a apuração dos saldos finais, e estas não foram finalizadas até a conclusão da nossa auditoria. Consequentemente, não nos foi possível concluir se eventual ajuste seria necessário nos ativos, passivos e no resultado do exercício como resultado da resolução deste assunto.” (original excerpt in Portuguese)
- [emphasis of matter] “Ênfase Continuidade operacional e debêntures a pagar a empresa relacionada Sem ressalvar nossa opinião, chamamos a atenção para o fato de que a Companhia incorreu em prejuízo líquido de R$ 33.866 e possui um passivo a descoberto (patrimônio líquido ativo) de R$ 414.127 mil em 31 de dezembro de 2013. Além disso, possui debêntures a pagar a empresa relacionada nos valores de R$ 960 mil e R$ 388.047 mil, demonstrados no circulante e não circulante, respectivamente bem como passivo tributário em processo de revisão conforme nota explicativa 9. Essas condições indicam a existência de incerteza significativa que pode levantar dúvida quanto à capacidade de liquidação saldo de debêntures a pagar e da continuidade operacional da Companhia. A continuidade operacional da Companhia depende do sucesso da administração na implementação de plano de reestruturação descrito na nota explicativa 2.” (original excerpt in Portuguese)

- Excerpt omitted; see the auditor's report in the document

Reference Form
Form of 2014, version 1, filed on 2014-05-30.
It is the only version available in the open data; earlier versions are not there.
- Related party (name omitted) · Coligada · Emissão de Debêntures · 389007000.00 · Em 31/12/2013 - R$ 389.007.000,00
- Related party (name omitted) · Coligada · Contrato de mútuo com prazo indeterminado · 261000.00 · em 31/12/2013 - R$ 261.000,00
- Related party (name omitted) · Coligada · Contrato de mútuo com prazo indeterminado. · 2966375.61 · Em 31/12/2013 - R$ 2.996.375,61

Limits of this section:
- In real documents, about 22% of the excerpts come out omitted by the name detector, including names of companies outside the registry.
- In tables, header lines (years, column titles) may come out as excerpts; the held-out cases had no tables with multi-line headers.

## How the engine was tested

In study 1, engine v1 detected 7 of 13 detectable cases, with 108.3 false alarms per detected case over the whole period and an alert rate of 49.9% among evaluated non-financial company-years (docs/summary.md).

## Limits

- No alert is not a clean bill of health. In Dechow, Sloan and Sweeney (1995), the most powerful accruals model rejected 'no manipulation' for only 28.1% of 32 companies sanctioned by the SEC (Table 7, p. 32).
- In study 1, 49.9% of evaluated non-financial company-years (48.3% including financial companies) were in alert, and the number of detections did not beat chance in a descriptive test done after the result: 7 detections in 13 detectable cases, against 7.37 expected by chance without assuming independent years (8.56 assuming independent years).
- Risk models use numbers that may be manipulated (Pereira and Martins 2015, p. 29). That is why the three families are not added up.

## Limitations

- Context indicators (Altman Z'' and Piotroski F) were not validated in Brazil by this project and do not trigger alerts

Evidence in the JSON: 38 DFP lines, each with document, version, filing date, account and location.

## Engine

motor v1 (estudo 1), branch estudo-1-final, commit 8c9a823; chiocca.v2 (produto, fases B a F1); leitor-f1-1 (commit c99c953). The generation date is in the demo manifest (demo/manifesto.json).

## Today's information (outside the reference date)

Registry status: ATIVO. Source: CVM registry of listed companies (cad_cia_aberta.csv), in the version downloaded for the project, not on the reference date.
Name in today's registry: HERCULES S/A - FABRICA DE TALHERES.